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Comparative linguistic analysis framework of human-written vs. machine-generated text

Lia Cornelia Culda^a, Raluca Andreea Nerişanu^a, Marian Pompiliu Cristescu^a, Dumitru Alexandru Mara^a, Adela Bâra^b and Simona-Vasilica Oprea^b

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ABSTRACT

Writing content with the assistance of artificial intelligence has increased in popularity and influenced every domain, yet the challenge of understanding the differences between machine-generated and human-authored writing remains. To address this issue, our research aims to propose a detailed framework for deciphering the characteristics of machine-written language by identifying distinct linguistic features, analyzing stylistic differences, evaluating sentiment consistency, and validating the performance of selected generative models in replicating human writing across different types of writing. Using free resources from Hugging Face, we selected three models: GPT-Neo 1.3B, Qwen2.5-1.5B-Instruct, and BloomZ-560M, and employed them without fine-tuning in a Google Colab environment using a consistent prompt. These models were used to generate introductions for corresponding human-authored texts. We compared the extracted features of 2,121 texts, using visualisation methods such as histograms, boxplots and Q-Q plots, alongside the Shapiro–Wilk test and the non-parametric Kruskal – Wallis test and Dunn’s post-hoc test. Our findings highlight both the advancements of open-source text-generating models and the persistent gaps in replicating the depth and richness of human writing, providing and contributing to the knowledge about artificial intelligence and its progress in the generative field, supporting the validation of open-source text-generative models through statistical comparisons.

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1. Introduction

The evolution of text-generative AI models has evolved from early rule-based systems to modern large language models (LLMs) with increasingly sophisticated capabilities. The accessibility of these models has accelerated progress, allowing developers to customise and fine-tune these models according to specific needs.

Zhang et al. emphasise the rapid rise of AI-generated content (AIGC) and its potential to transform industries like education, gaming, advertising, and code development (Zhang et al., 2023). However, Ma et al. note gaps in-depth and quality, as AI-generated texts lack

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the personal experiences, emotional nuances, and stylistic diversity of human writing (Ma et al., 2023). Human texts contain personal experiences, emotional nuances and stylistic diversity that increase engagement and recognition. AI-generated texts, on the other hand, are informative but lack human touch and linguistic complexity (Jones, 2022; Kojima, 2024).

Wikipedia, the largest collaborative multilingual encyclopedia, is a rich resource for natural language processing research (Mehdi et al., 2017; Sugandhika & Ahangama, 2022). Similarly, blog articles with different styles serve as an excellent source of human-authored texts for training and evaluating generative LLMs (Boas & Boas, 2023; Radensky et al., 2024; Thomas et al., 2024). Also, AI models can generate scientific abstracts that resemble human writing, but they are often associated with lower perceived quality. (Howard et al., 2024)

In this research, we statistically assess the differences between machine-generated text from publicly available generative pre-trained models and human-authored text. We used free resources from Hugging Face. We selected three small models and deployed them without fine-tuning in a Google Colab environment using a consistent prompt. We also selected three datasets of text introductions from Wikipedia, blog articles and scientific abstracts, representing formal, informal and academic writing styles and tonalities. The total number of human-written texts reached 528 records, of which 200 were from Wikipedia, 128 from blog articles and 200 from academic abstracts. In addition, 1,584 texts were generated by the AI, with each model generating 528 outputs for the corresponding text source, resulting in a total of 8,560 generated sentences. We calculate various text features, including lexical, syntactic, readability, and stylistic attributes, as well as perform sentiment analysis on the generated text. Our lexical metrics included word count, number of unique words, type-token ratio (TTR), average word length, and hapax legomena (Ali & Hussein, 2014; Churunina et al., 2023; Fergadiotis et al., 2013; Lytvyn et al., 2018; Ross, 1977). Syntactic complexity was measured by the number of sentences, average sentence length and part-of-speech frequencies (Churunina et al., 2023; Fergadiotis et al., 2013; Khany & Kafshgar, 2016; Lytvyn et al., 2018). We assessed readability using Flesch Reading Ease and Flesch-Kincaid Grade Level (Crossley et al., 2008; Ludbrook, 2006; Talburt, 1986). Stylistic features included the proportion of passive voice, function words, adverbs and adjectives (Anggraini, 2018; Kettunen, 2014; Ross, 1977; Sharma et al., 2015). Sentiment analysis captured polarity and subjectivity (Dehkharghani et al., 2016; Kumawat et al., 2021; Priyadarshana et al., 2015). By comparing these features with those of human-written text, we seek to understand the current capabilities and limitations of publicly available generative pre-trained models.

Our focus on open-source resources highlights the potential for anyone to experiment with and apply generative AI, promoting wider accessibility and trust (Candel et al., 2023). By statistically examining the differences between AI – and human-generated texts, our research highlights the current capabilities of publicly available generative models. The findings may inform future improvements to AI models and highlight specific areas where machine-generated text still differs from human authorship.

2. Literature review

In this section, we review previous studies on text-generative AI models and explore how text features have been analyzed in the existing literature. A brief summary of this review can be found in Table 1.

Table 1. Brief description of the previous studies on objectives, methods and findings of AI texts.

Reference	Objectives	Methods	Findings
(Ma et al., 2023)	Examine the gap in depth and quality between AI-generated scientific content and human-written material.	Comparison of scientific texts generated by AI and human authors.	AI lacks depth compared to human authorship in scientific content.
(Candel et al., 2023)	Showcase the importance of open-source resources for democratizing AI.	Literature review on open-source AI development.	Open-source models democratise AI and enhance accessibility.
(Gozalo-Brizuela & Garrido-Merchan, 2023)	Discuss the advancements of multimodal models in AI.	Analysis of multimodal AI models and their applications.	Multimodal models extend AI's capacity beyond text.
(Anantrasirichai & Bull, 2022)	Explore AI's role in enhancing creativity in the content production process.	Review of AI tools in creative processes, e.g. GANs.	AI aids human creativity rather than replacing it.
(Chakraborty et al., 2023)	Examine the quality of text generated by advanced LLMs, focusing on coherence and contextual relevance.	Empirical analysis of AI model outputs in tasks like poetry and essays.	LLMs generate high-quality, fluent, and coherent text.
(Cheng et al., 2023)	Evaluate the reliability of AI-generated scientific abstracts.	Content analysis of AI-generated scientific abstracts.	AI-generated scientific abstracts often have factual inaccuracies.
(Grimaldi & Ehrler, 2023)	Analyze the lack of fresh perspectives in AI-generated content.	Case study analysis of human creativity vs. AI.	AI content lacks subjective and fresh perspectives.
(Herbold et al., 2023)	Compare lexical diversity and structure in AI-generated essays.	Text analysis of AI-generated vs. human-written essays.	AI essays exhibit high lexical diversity but lack depth.
(Muñoz-Ortiz et al., 2023)	Analyze sentence complexity and emotional intensity differences between AI and human texts.	Comparative analysis of syntactic features and sentence lengths.	Human texts show more complex sentence structures and emotional intensity.
(Reviriego et al., 2023)	Compare lexical diversity in AI-generated vs. human-written texts.	Lexical diversity analysis using unique words and vocabulary.	AI-generated text has lower lexical diversity.
(Liu et al., 2023)	Enhance AI-detection accuracy by comparing syntactic structures in AI and human texts.	Syntactic structure analysis and model training for detection.	Detection models achieve high accuracy distinguishing AI content.
(Vanmassenhove et al., 2019)	Discuss lexical richness limitations in machine translation models.	Comparative study on lexical choices in machine translation.	Machine translation lacks lexical richness of human translations.
(Shah et al., 2023)	Identify stylistic features that help distinguish AI-generated text.	Use of stylistic features for AI detection models.	Model accuracy reaches 93% using stylistic features for AI detection.
(Imperial & Madabushi, 2023)	Examine the readability of texts generated by various AI models.	Readability analysis based on sentence structure and vocabulary.	AI texts are more readable but simpler in structure.
(Schouten et al., 2016)	Enhance sentiment analysis by using hybrid approaches to classify AI and human texts.	Hybrid approach integrating lexical and syntactic features.	Hybrid sentiment analysis improves classification accuracy.

Generative AI models, especially LLMs, have advanced significantly. Multimodal models like DALL-E and CLIP extend LLMs by integrating data types such as images, enabling tasks like text-to-image generation. These innovations build on the transition from statistical to neural models, culminating in transformer-based PLMs (Gozalo-Brizuela & Garrido-Merchan, 2023; Zhao et al., 2023). AIGC is being used across industries, particularly in creative fields, where AI tools like GANs support content creation, post-production, and compression, enhancing human creativity rather than replacing it (Anantrasirichai & Bull, 2022). Despite their capabilities, LLMs face challenges like hardware requirements, security, and customisation (Zhao et al., 2023).

The quality of AI-generated texts from LLMs such as GPT-3.5 and GPT-4 has reached a level where it can hardly be distinguished from texts written by humans. This is particularly evident in tasks such as poetry, essays and news generation, where the models produce fluent, coherent and contextual content (Chakraborty et al., 2023).

However, AI-generated text has its limitations. It can lack human creativity, nuanced understanding and originality. It is also prone to factual inaccuracies, redundancy and coherence problems (Cheng et al., 2023; Ma et al., 2023).

Furthermore, AI tools often rely on pre-existing data, which may result in a lack of fresh perspectives, unlike human writers who may integrate lived experiences and subjective analysis into their work (Grimaldi & Ehrler, 2023). Although AI-generated texts are useful for tasks such as summaries and drafts, they may lack the depth and epistemic markers of human scientific discourse (Herbold et al., 2023).

Human-authored texts exhibit significant variability, ranging from simple to complex structures, influenced by linguistic, cognitive, and contextual factors. For instance, studies analyzing Chinua Achebe's "Things Fall Apart" (Babiker, 2023) and individual writing styles (Verma & Srinivasan, 2019) demonstrate that human writing incorporates diverse lexical categories and complex grammatical structures, contributing to engaging and vivid narratives. Additionally, research on writing quality highlights that syntactic complexity and lexical sophistication are indicators of high-quality writing (Maamuujav et al., 2021).

Machine-generated text often has less variance in sentence length and relies on simpler structures compared to human texts, which show more intricate dependencies and emotional intensity. Studies highlight differences in sentence-length distributions, readability, and syntactic complexity (Gamon, 2004; Muñoz-Ortiz et al., 2023; Watkins & Knapp, 2005). For instance, Reviriego et al. found that AI-generated texts, like those from GPT models, exhibit lower lexical diversity and fewer unique words than human texts (Reviriego et al., 2023). AI-generated essays often show complex syntax, while human essays demonstrate the greater lexical richness, aiding detection models like RoBERTa to achieve over 90% accuracy (Liu et al., 2023). Similarly, Vanmassenhove et al. noted machine translation systems prioritise frequent word patterns over lexical richness, increasing repetition (Vanmassenhove et al., 2019).

Stylistic features such as syllable count, word length, sentence structure and punctuation ratios have been used to classify texts as AI-generated or human-written with 93% accuracy (Shah et al., 2023). For example, a study analyzing machine-generated texts from various models found that models like GPT tend to use simpler sentence structures and vocabulary, resulting in higher readability scores (Imperial & Madabushi, 2023).

Sentiment analysis is also important to distinguish between AI-generated and human-written texts. A hybrid approach to extracting sentiment from texts using different patterns,

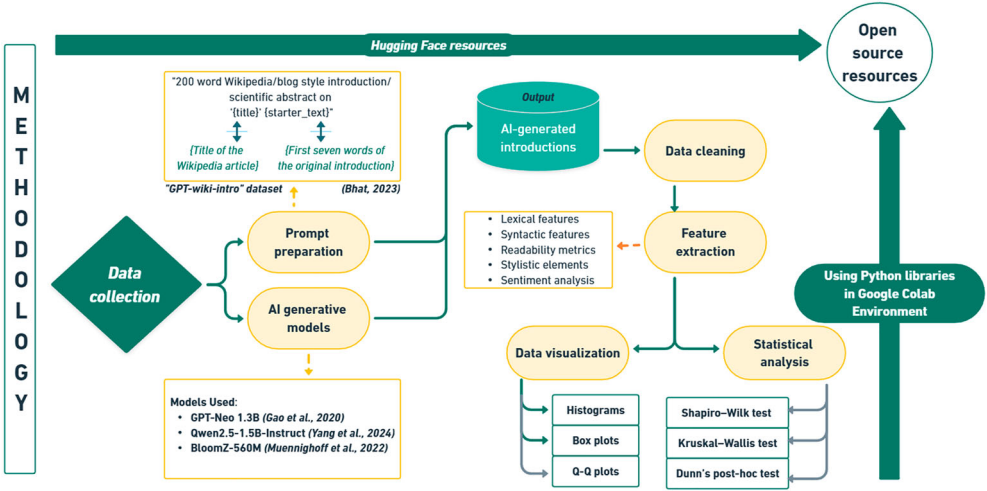


Figure 1. Proposed methodology with a focus on open-source resources.

such as lexical and syntactic features, has improved the accuracy of sentiment analysis by revealing subtle differences in the text (Schouten et al., 2016).

3. Methodology

Our methodology provides an extensible framework for data processing and analysis that facilitates the replication of this study and its adaptation to different contexts. All procedures were implemented using Python in a Google Colab environment, leveraging publicly available libraries. This open-source approach, as illustrated in Figure 1, ensures both accessibility and reproducibility.

The framework follows a systematic sequence of steps:

3.1. Step 1. Dataset and text generation

The dataset, D is compiled from three distinct sources, each of which was utilised to extract records in a direct and random manner. These sources are publicly available on Hugging Face and include: D_W - a collection of Wikipedia articles from Bhat's "GPT-wiki-intro" dataset (Revision 0e458f5) (Bhat, 2023), D_B - a corpus of blog articles from the Medium platform (2016–2019) ([dataset]) and D_A - scientific abstracts of arXiv articles up to the end of 2021 (Tian et al., 2023). The total number of human-written texts in D reached $N_h = 528$ records, of which $N_w = 200$ were from D_W , $N_b = 128$ from D_b and $N_a = 200$ from D_a . These are used to create human-authored and AI-generated text samples:

$$D = \{D_w, D_b, D_a\} \quad (1)$$

For the generation of AI-authored introductions, we select three publicly available text generative models, M_j , available on Hugging Face: M_1 : EleutherAI's GPT-Neo 1.3B (Black et al., 2021; Gao et al., 2020), M_2 : Qwen2.5-1.5B-Instruct (Yang et al., 2024) and M_3 : BigScience's BloomZ-560M (Muennighoff et al., 2022).

We chose to use freely available tools in a Google Colab environment, which provides us with the necessary computational resources for our research. The GPT-Neo 1.3B model, developed by EleutherAI, is a 1.3-billion-parameter model recognised for its robust performance on language modelling tasks (Black et al., 2021; Gao et al., 2020). The Qwen2.5-1.5B-Instruct model is an instruction-tuned variant designed to effectively follow human instructions, enhancing its utility in generating coherent and contextually appropriate text (Yang et al., 2024). The BloomZ-560M model, created by BigScience, is a multilingual model trained for cross-lingual generalisation through multitask fine-tuning, offering versatility in handling diverse linguistic inputs (Muennighoff et al., 2022).

To generate AI introductions, we employed a prompt structure similar to Bhat's methodology (Bhat, 2023).

Each AI model generates a text sample $A_w^{M_j}, A_b^{M_j}, A_a^{M_j}$ for the corresponding w, b, a articles in the subsets of D , using the following prompt:

Prompt : "200 word Wikipedia style introduction on{title}{starter_text}" (2)

"200 word blog style introduction on{title}{starter_text}" (3)

"200 word scientific abstract on{title}{starter_text}" (4)

where: *title* – Title of the article *and starter_text* – First seven words of the human-authored introduction.

Before running the prompts, the selected Wikipedia and scientific data had already been cleaned. However, additional pre-processing was required for the blog articles. We removed extra spaces, excluded articles that started with the structured format "by author's name" or contained less than 30 words, and kept only the first 300 words (or less if the introduction was shorter) to account for the approximate length of Wikipedia introductions. Emojis and special characters were also removed. These cleaning steps ensured that the sets of human-authored introductions were comparable in length, style, and clarity.

The prompts provide the AI models with a contextual anchor derived from the existing content and help to maintain the coherence and continuity of the generated text (Bhat, 2023). The consistency of the prompt across all generations allowed for minimising the variability caused by different input structures.

Our dataset now consists of 3 groups, G , each with 4 sources, for every source in D there are in addition three other sources with the results generated by each model for the corresponding source:

$$G_w = \{S_{(w)} \mid w \in D_w\}, \text{ where } S_{(w)} = \{w, A_w^{M_1}, A_w^{M_2}, A_w^{M_3}\} \quad (5)$$

$$G_b = \{S_{(b)} \mid b \in D_b\}, \text{ where } S_{(b)} = \{b, A_b^{M_1}, A_b^{M_2}, A_b^{M_3}\} \quad (6)$$

$$G_a = \{S_{(a)} \mid a \in D_a\}, \text{ where } S_{(a)} = \{a, A_a^{M_1}, A_a^{M_2}, A_a^{M_3}\} \quad (7)$$

3.2. Step 2. Generated text cleaning

Once the texts were generated, a thorough data clean-up was carried out to improve quality and consistency. Python's "re" module was used extensively for this process. For instance, URLs were removed using patterns like `re.sub(r'http$+|www$+|https$+', '', text)`, while

Table 2. Synthesised measured features and studies that justify the selection of features for text analysis (see Glossary).

Feature	Metrics	Studies
Lexical (f_l)	word count (l_1) unique word count (l_2) type-token ratio (TTR) (l_3) average word length (l_4) hapax legomena (l_5)	(Ali & Hussein, 2014; Churunina et al., 2023; Fergadiotis et al., 2013; Lytvyn et al., 2018; Ross, 1977)
Syntactic (f_s)	sentence count (s_1) average sentence length (s_2) counts of nouns (s_3), verbs (s_4), adjectives (s_5), and adverbs (s_6)	(Churunina et al., 2023; Fergadiotis et al., 2013; Khany & Kafshgar, 2016; Lytvyn et al., 2018)
Readability (f_r)	Flesch Reading Ease (r_1) Flesch-Kincaid Grade Level (r_2)	(Crossley et al., 2008; Ludbrook, 2006; Talbert, 1986)
Stylistic (f_y)	passive voice percentage (y_1) function word percentage (y_2) adverb percentage (y_3) adjective percentage (y_4)	(Anggraini, 2018; Kettunen, 2014; Ross, 1977; Sharma et al., 2015)
Sentiment analysis (f_a)	sentiment polarity (a_1) sentiment subjectivity (a_2)	(Dehkharghani et al., 2016; Kumawat et al., 2021; Priyadarshana et al., 2015)

unusual characters were filtered out with `re.sub(r'[^\A-Za-z0-9\s.,!?"\''-]', "", text)`. Additional spaces were condensed using `re.sub(r'\s+', ' ', text).strip()`.

To address incomplete sentences, the text was truncated to its last valid sentence using built-in Python methods such as `text.endswith('.')` and `text.rfind('.')`. For example, a model generated the following introduction: “Born into a noble family, her upbringing allowed her access to various educational opportunities. However, due to societal norms at the time”, truncation ensured coherence: “Born into a noble family, her upbringing allowed her access to various educational opportunities.”.

Additionally, rows with empty values (NaN) were removed to refine the dataset. While each model generated responses for Wikipedia-style introductions and scientific abstracts, inconsistencies were observed in blog-style outputs. Specifically, the Bloomz model generated responses for only 128 out of 200 prompts.

Define $\text{Clean}(T)$ as a cleaning function applied to text T generated by $A_w^{M_j}$, $A_b^{M_j}$, $A_a^{M_j}$ to ensure quality and coherence:

$$\text{Clean}(T) = T - \left\{ \begin{array}{l} \text{control characters, URLs, unusual characters,} \\ \text{extra spaces, incomplete sentences, empty rows} \end{array} \right\} \quad (8)$$

Apply $\text{Clean}(T)$ results in $N_t = N_w + N_b + N_a = 1584$ generated articles introductions

3.3. Step 3. Feature extraction

After cleaning, five categories of text features – lexical, syntactic, reading, stylistic and sentiment – were extracted to allow detailed quantitative evaluation (see Table 2).

Let F represent the set of features extracted from each text sample.

$$F = \{f_l, f_s, f_r, f_y, f_a\} \text{ or } F = \{l_1, \dots, l_5, s_1, \dots, s_6, r_1, r_2, y_1, \dots, y_4, a_1, a_2\} \quad (9)$$

Define the feature extraction function:

$$\text{ExtractFeatures}(T) = \{f_l(T), f_s(T), f_r(T), f_y(T), f_a(T)\} \quad (10)$$

We computed F_{Gw} , F_{Gb} , F_{Ga} , for $N = N_h + N_t = 2112$ texts, categorised into three distinct corpora. We extracted features for 800 Wikipedia texts, comprising 200 texts authored by

humans and 600 AI-generated texts (200 per selected generative model). Additionally, 512 blog articles were used to extract features, consisting of 128 human-authored texts and 384 AI-generated texts (128 per model). Lastly, we used 800 scientific abstracts to extract textual features, of which 200 were authored by humans and 600 were AI-generated (200 per selected generative model).

3.4. Step 4. Statistical assessment

We used visualisations to facilitate the comparison between human-authored and AI-generated texts, providing insights into their structural and statistical differences.

Let $V(F)$ represent visualisation functions for feature distributions:

$$V(F) = \{\text{histograms, boxplots, } Q - Q \text{ plots}\} \quad (11)$$

In addition to the visualisations that provide a basis for assessing the normality of the data, we use the Shapiro – Wilk test to determine whether the trait values in our data set follow a normal distribution. Our null hypothesis is H_0 : *Feature values follow a normal distribution*. Our hypothesis is rejected if $p < \alpha$, where $\alpha = 0.05$.

3.4.1. Step 4.1 Group comparison

We employ the Kruskal – Wallis test to assess whether statistically significant differences exist among the groups, without relying on the assumption of normality. To complement this analysis, we calculate epsilon squared as a measure of effect size, providing insight into the magnitude of these differences. If the Kruskal – Wallis test yields a significant result, we proceed with Dunn’s post-hoc test, applying the Bonferroni correction to account for multiple comparisons and ensure statistical validity. This approach allows us not only to identify the presence of differences but also to determine precisely where they occur. Our null hypothesis is H_0 : *No significant difference between human-authored and AI-generated features*. Our hypothesis is rejected if $p < \alpha$, where $\alpha = 0.05$.

3.5. Step 5. Evaluation metrics

Define the evaluation metric functions:

$$Eval(F_{Gw}) \text{ and } Eval(F_{Gb}) \text{ and } Eval(F_{Ga}) \quad (12)$$

These functions present the results of all statistical tests and visual comparisons for each group source F , assessing the differences between human-authored texts and their AI-generated counterparts across various models.

4. Results

4.1. Histograms

4.1.1. Histograms for lexical features

As shown in the Figure 2, across the analyzed text sources, machine-generated outputs tend to be more concise, with fewer word counts and a narrower range of unique words.

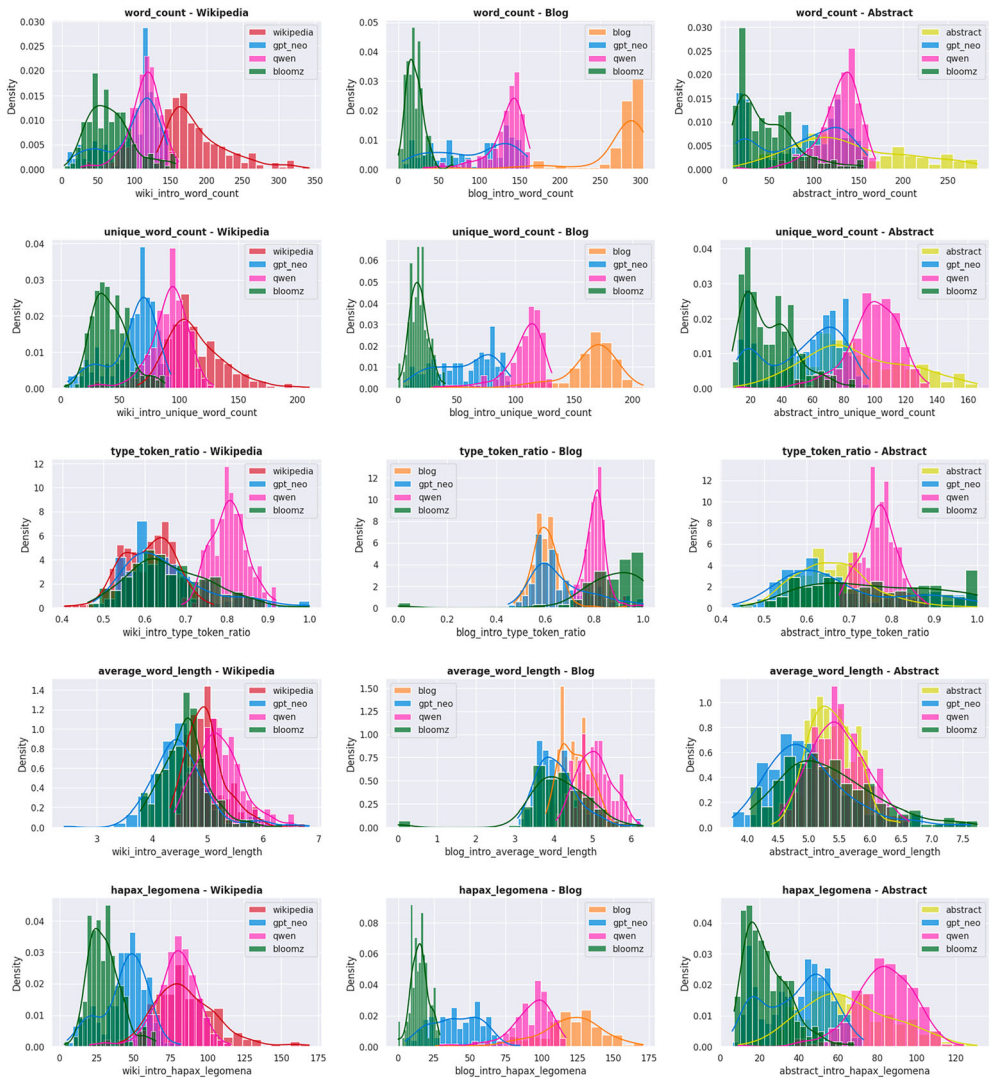


Figure 2. Distribution of lexical features across machine-generated and human-authored texts.

Also, they exhibit lower lexical diversity, as indicated by their type-token ratios and the frequency of hapax legomena. In comparison, original Wikipedia introductions, blog posts, and academic abstracts display greater linguistic complexity.

Academic abstracts show the most significant disparity. They typically contain more specialised vocabulary and longer average word lengths than machine-generated texts, highlighting the limitations of these models in replicating highly technical writing. Blog posts, while naturally diverse, also surpass GPT outputs in their use of unique words and overall lexical richness. Wikipedia texts fall between these extremes, yet even in this case, machine-generated texts tend to be simpler and more uniform.

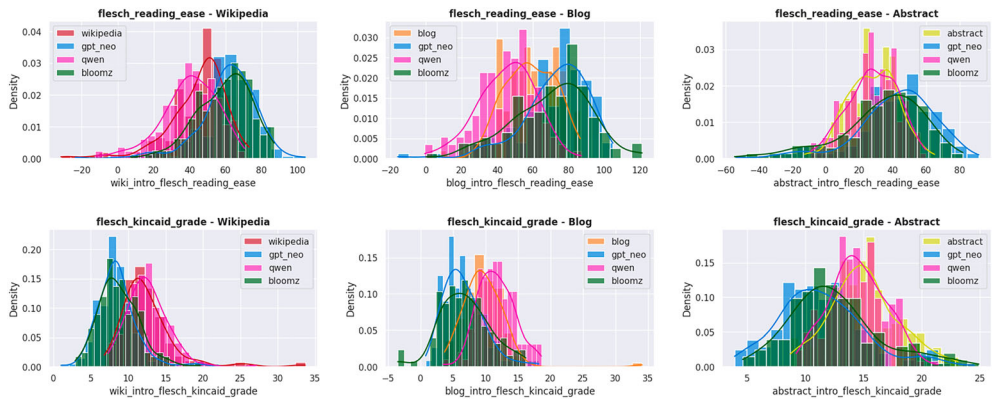


Figure 3. Distribution of readability metrics features across across machine-generated and human-authored texts.

4.1.2. Histograms for readability metrics

As shown in Figure 3, in all source groups examined – Wikipedia, blogs and academic introductions – texts generated with GPT models tend to be more readable than their original counterparts. This trend is reflected in the consistently higher Flesch Reading Ease scores and lower Flesch-Kincaid Grade Levels, indicating simpler language and sentence structures.

The most significant difference is observed in academic abstracts, where the AI-generated content is much more accessible than the often complex and technical original texts. In contrast, the difference is less pronounced for blog posts, as these are already designed for broad readability. Wikipedia content lies between these two extremes, with the GPT-generated versions showing slightly improved readability compared to the original articles.

4.1.3. Histograms for stylistic features

As shown in Figure 4, across different human-authored text source, machine-generated texts tend to exhibit lower usage of passive voice and fewer adverbs and adjectives compared to human-written texts within the same group. Academic abstracts demonstrate a stronger reliance on passive constructions and function words, whereas blog posts are characterised by a higher frequency of adverbs and adjectives. Wikipedia articles generally occupy an intermediate position between these two groups. In contrast, machine-generated outputs appear more uniform across groups, displaying lower rates of passive voice usage and a reduced presence of descriptive elements.

4.1.4. Histograms for syntactic features

As shown in Figure 5, across all three human-authored text sources, machine-generated texts generally contain fewer sentences and exhibit a shorter average sentence length compared to human-written originals. This pattern is reflected in both the total sentence count and the syntactic complexity of the texts. Additionally, machine-generated content tends to employ a more limited range of linguistic features, using fewer nouns, verbs, adjectives, and adverbs, which contributes to a simpler overall structure.

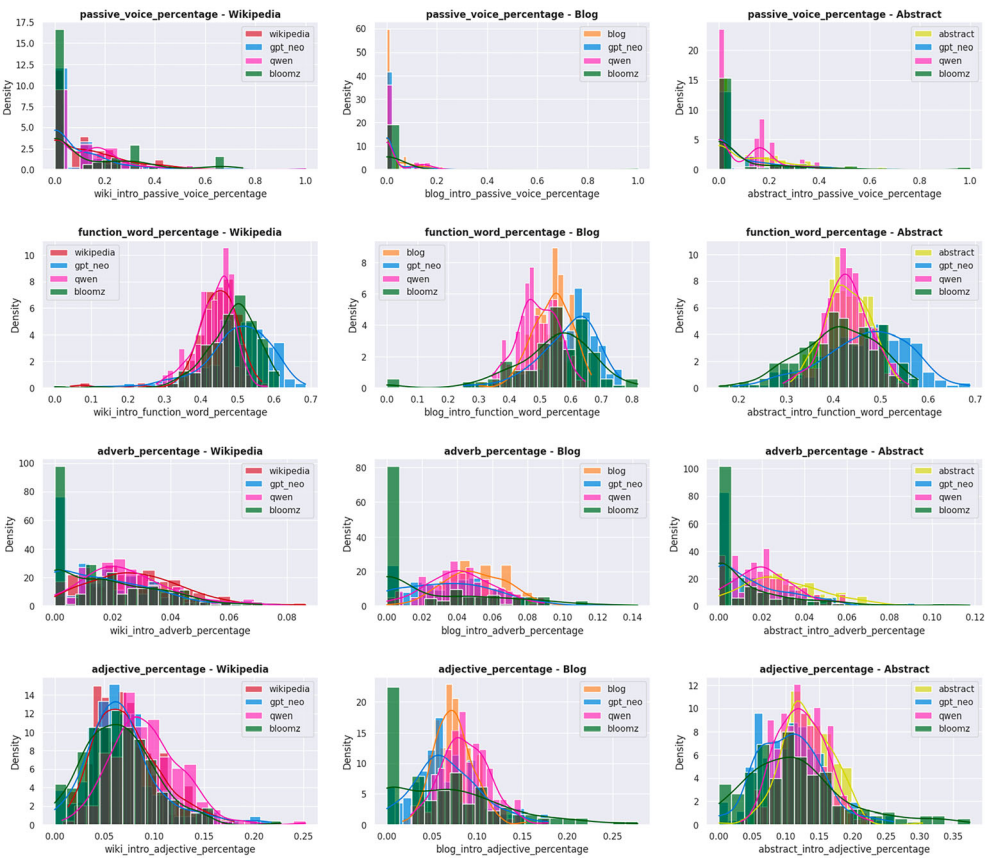


Figure 4. Distribution of stylistic features across machine-generated and human-authored texts.

The differences between human-written and AI-generated texts are particularly pronounced in academic abstracts. These texts typically contain dense, technical language, which the GPT models fail to replicate at the same level of complexity. In contrast, blog posts, which are naturally less formal, still demonstrate greater linguistic diversity and more varied use of parts of speech than the corresponding generated versions.

4.1.5. Histograms for sentiment analysis

As shown in Figure 6, across all three groups analysed, texts generated by the chosen models tend towards more positive sentiment polarity compared to their human-written counterparts. This is more evident in blog posts and abstract content. In terms of subjectivity, variations occur depending on the text type: while machine-generated blog posts often match or slightly exceed the subjectivity levels of human-authored texts, outputs for Wikipedia introductions and abstracts sometimes demonstrate lower subjectivity than their equivalents. Overall, machine-generated texts generally convey a more positive and approachable tone, contrasting with the more neutral or varied sentiment observed in human-written content.

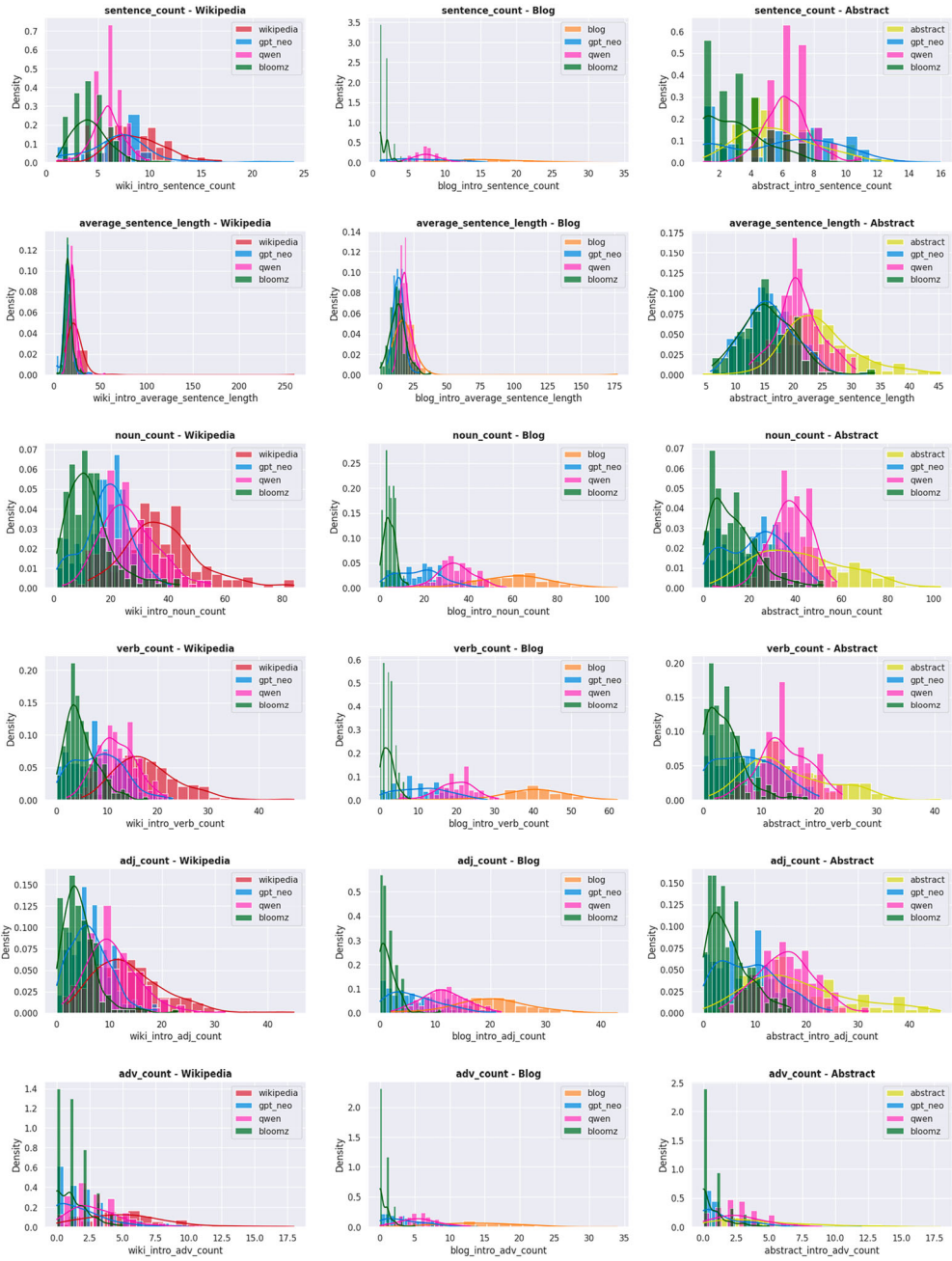


Figure 5. Distribution of syntactic features across machine-generated and human-authored texts.

4.2. Box plots

4.2.1. Box plot for lexical features

As shown in Figure 7, the texts generated exhibit consistently lower median values and narrower ranges in terms of word count, unique word count, and lexical diversity measures,

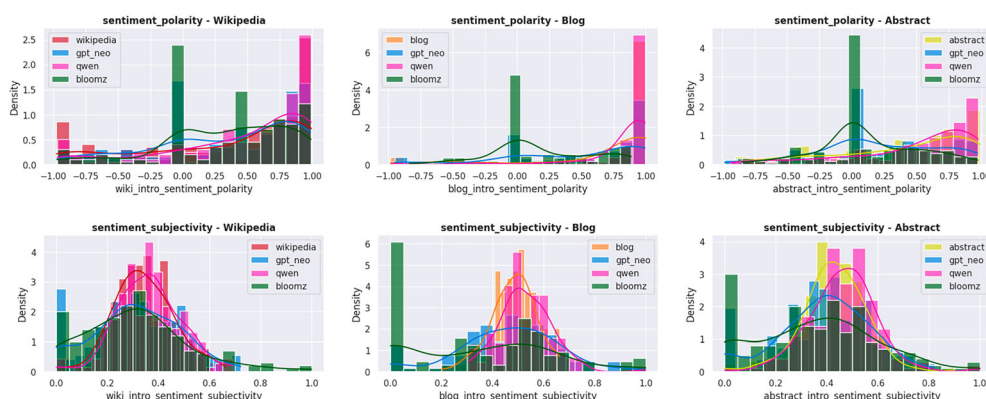


Figure 6. Sentiment polarity and subjectivity of machine-generated and human-authored texts.

such as the type-token ratio and hapax legomena, compared to human-written texts. This pattern is evident across all analysed sources – Wikipedia, blog posts, and academic abstracts – where machine-generated content tends to cluster around simpler and more uniform language structures. In contrast, human-authored texts display a wider distribution of lexical features, suggesting variation in writing style and complexity.

4.2.2. Box plot for readability metrics

As shown in Figure 8, machine-generated texts show higher Flesch Reading Ease scores and lower Flesch-Kincaid Grade Levels compared to corresponding human-authored text, suggesting that generated texts are more accessible and easier to comprehend, particularly for academic abstracts. Additionally, the narrower interquartile ranges indicate reduced variability in complexity across generated outputs, demonstrating a more consistent writing style. These trends align with the patterns observed in the histograms, reinforcing that GPT models enhance readability while maintaining a uniform and simplified textual structure.

4.2.3. Box plot for stylistic features

As shown in Figure 9, machine-generated texts tend to be simpler and more direct, exhibiting less stylistic variation compared to human writing of each source, as evidenced also by the histograms of stylistic features. GPT models demonstrate consistent linguistic patterns across various formats, including Wikipedia, blogs, and academic abstracts. In contrast, human-authored texts exhibit greater variability, especially in academic writing, which often employs more passive voice and formal language.

4.2.4. Box plot for syntactic features

As shown in Figure 10, machine-generated texts are more concise and uniform, with fewer sentences, shorter sentence lengths, and lower word variety compared to human-written texts. Their distributions cluster around smaller values with relatively tight interquartile ranges. In contrast, human-authored texts, especially academic abstracts, demonstrate greater linguistic complexity and variability. Blogs and Wikipedia articles fall in between, showing more diversity than AI-generated texts but less formal structure than academic writing.

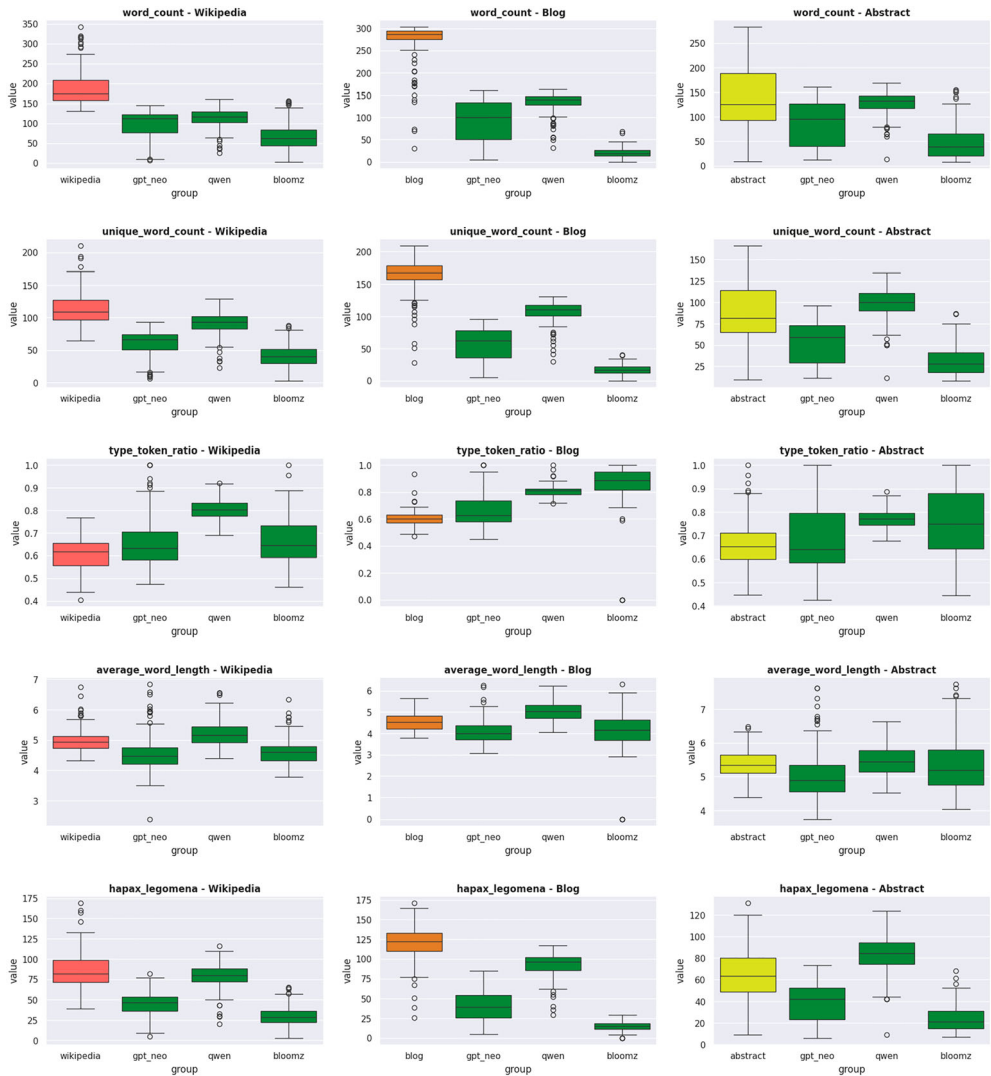


Figure 7. Box plot displaying the distribution of lexical features across machine-generated text and human-authored texts.

4.2.5. Box plot for sentiment analysis

As shown in Figure 11, machine-generated texts exhibit consistently positive tone and higher subjectivity than human-written content, with narrower variability. In contrast, human-authored texts, especially in academic and Wikipedia domains, tend to be more neutral, formal, and diverse in sentiment.

4.3. Q-Q plots

We created Q-Q plots to visually verify data normal distribution, with a line indicating the normal distribution's location for each text group.

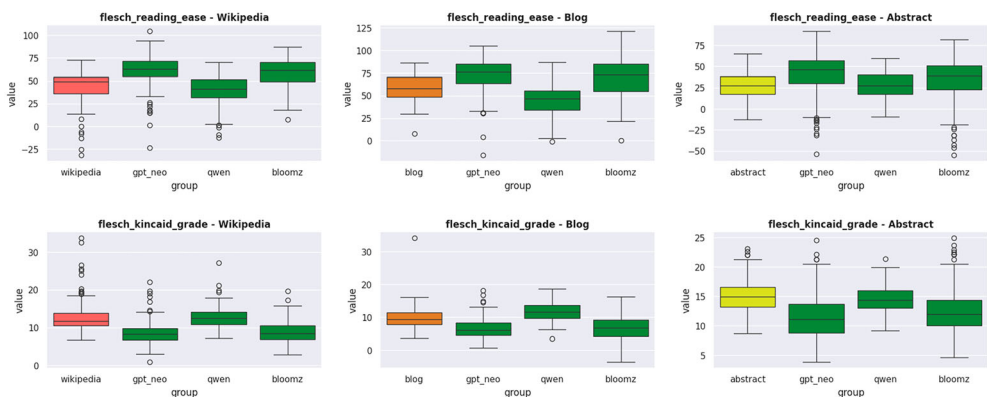


Figure 8. Box plot displaying the distribution of readability metrics across machine-generated and human-authored texts.

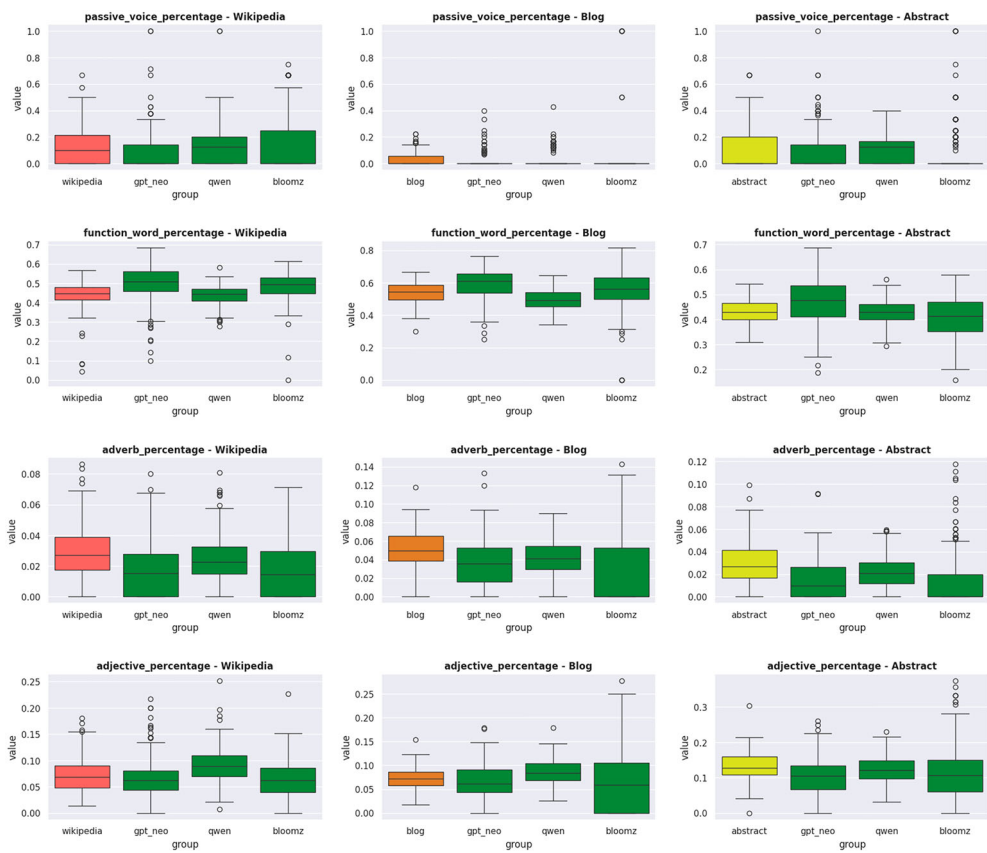


Figure 9. Box plot displaying the distribution of stylistic features across machine-generated and human-authored texts.

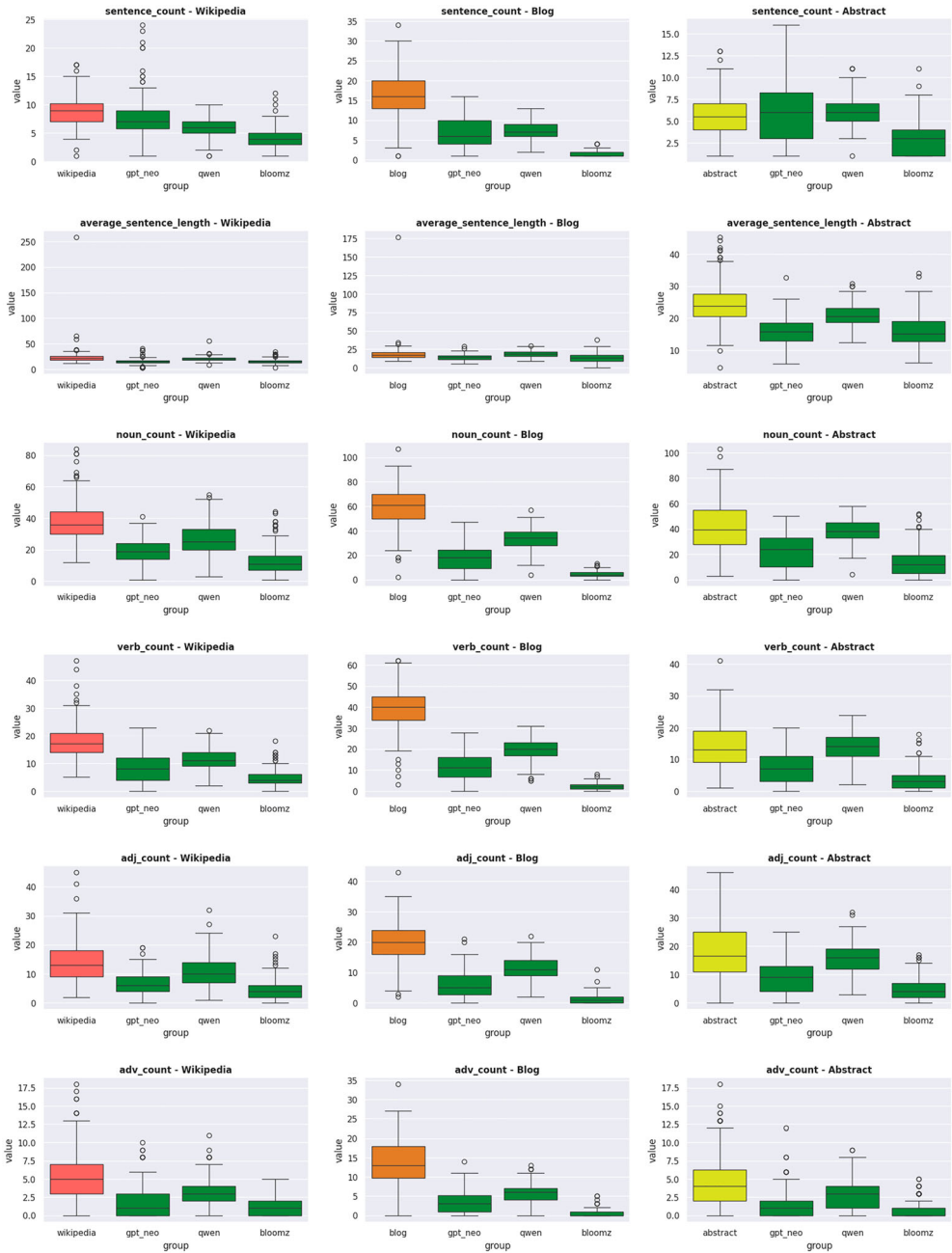


Figure 10. Box plot displaying the distribution of syntactic features across machine-generated and human-authored texts.

4.3.1. Q-Q plot for lexical features

As shown in Figure 12, none of the distributions align perfectly with a normal distribution. The machine-generated texts tend to fall closer to the diagonal, indicating a reduced occurrence of extreme values; however, they still show some deviations, especially in the tails. In

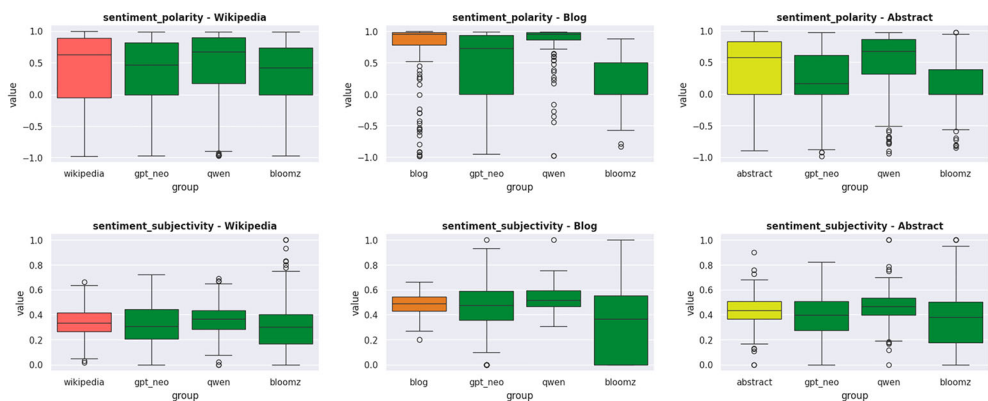


Figure 11. Box plot displaying the sentiment polarity and subjectivity across machine-generated and human-authored texts.

contrast, the human-written texts showcase more significant departures from normality, characterised by heavier tails and greater overall variability.

4.3.2. Q-Q plot for readability metrics

As shown in Figure 13, the distributions of reading ease and grade level do not fully align with the theoretical normal distribution. However, machine-generated texts tend to exhibit a tighter clustering and a preference for higher reading ease values. Conversely, human-generated content, particularly academic abstracts, displays a greater degree of deviation and variability, suggesting a wider range of linguistic complexity.

4.3.3. Q-Q plot for stylistic features

As shown in Figure 14, none of the linguistic metrics perfectly follow a normal distribution, but the machine-generated texts generally cluster more tightly, indicating more uniform usage.

4.3.4. Q-Q plot for syntactic features

As shown in Figure 15, none of the distributions for sentence count, average sentence length, or part-of-speech counts align with a normal distribution. The texts generated by GPT-Neo, Qwen, and Bloomz tend to cluster more closely along a narrower section of the diagonal, suggesting a more uniform style with fewer extreme values. In comparison, the human-written texts from Wikipedia, blogs, and abstracts demonstrate a greater degree of variability, as seen in their deviation from the reference line.

4.3.5. Q-Q plot for sentiment analysis

As shown in Figure 16, deviations from normality are observed in sentiment polarity and subjectivity, with uniform mild skewness. Partial alignment in some regions suggests localised normality, consistent with box plot and histogram patterns.

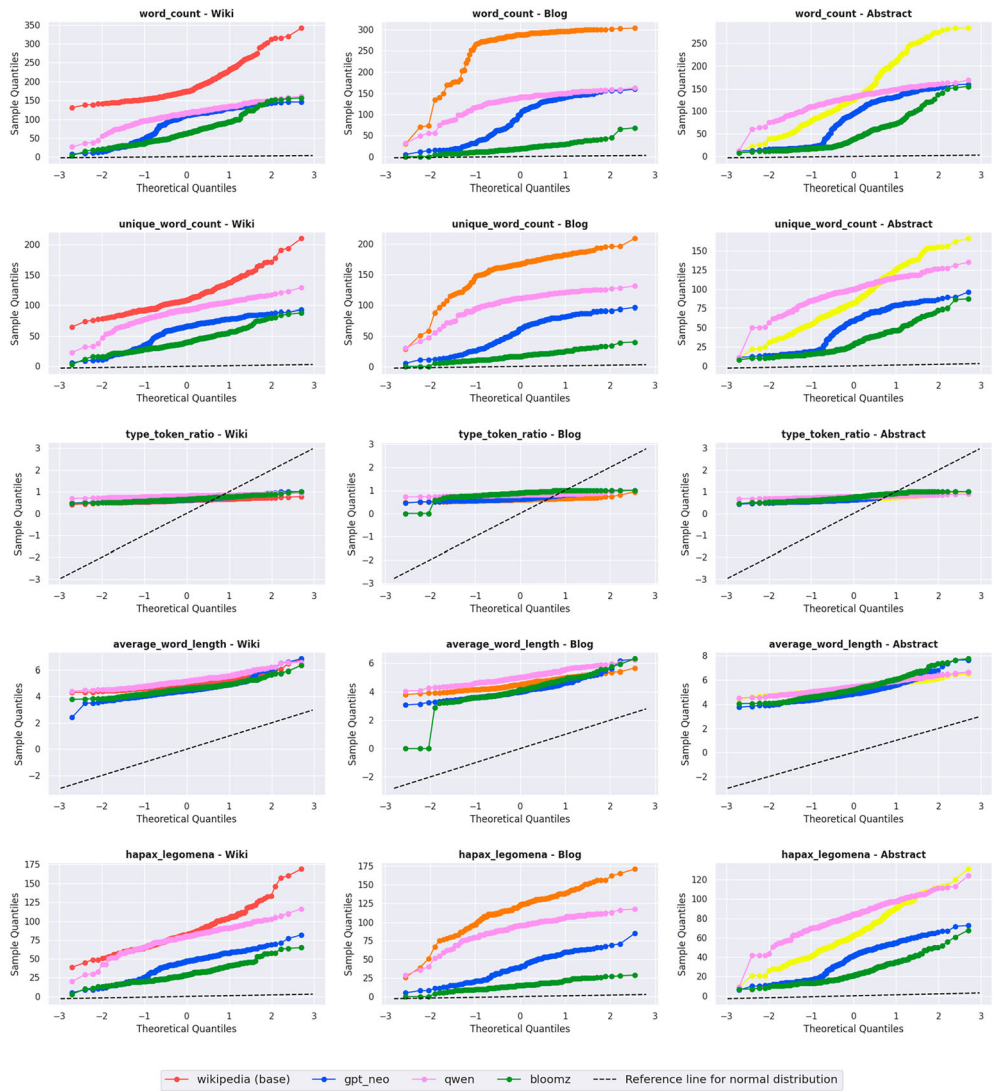


Figure 12. Q-Q plot displaying the distribution of lexical features across machine-generated and human-authored texts.

4.4. Normality test

We used the Shapiro–Wilk Test to confirm Q-Q plot findings on data normality.

As shown in Figure 17, most features deviate from normality ($p < 0.05$), consistent with Q-Q plots. Among the datasets, Wikipedia introductions display normality in TTR and sentiment subjectivity, while Qwen introductions exhibit normality in TTR, verb count, and sentiment subjectivity.

As shown in Figure 18, most features deviate significantly from normality. However, for blog and Qwen intros, features like noun and adverb counts, adjective percentages, function word percentages, and sentiment subjectivity show higher p -values, suggesting

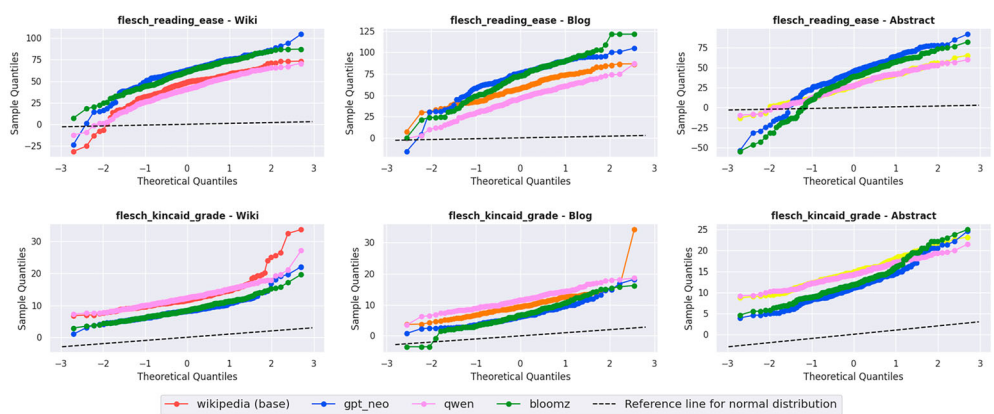


Figure 13. Q-Q plot displaying the distribution of readability metrics across machine-generated and human-authored texts.

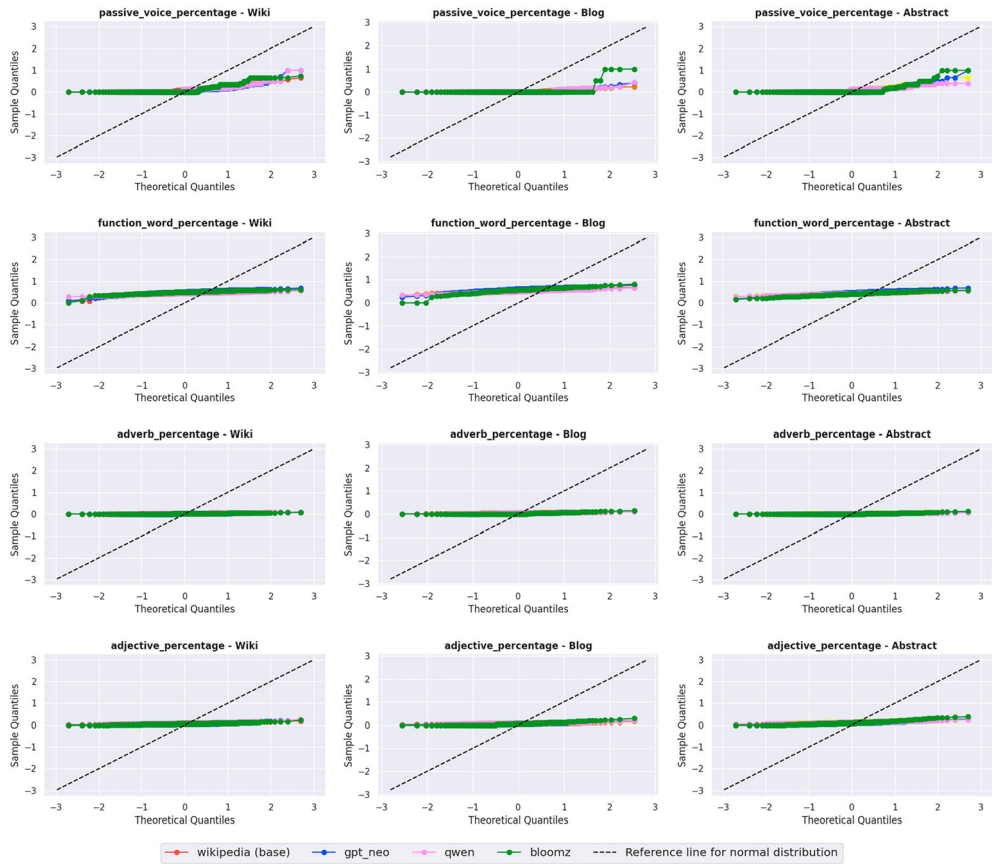


Figure 14. Q-Q plot displaying the distribution of stylistic features across machine-generated and human-authored texts.

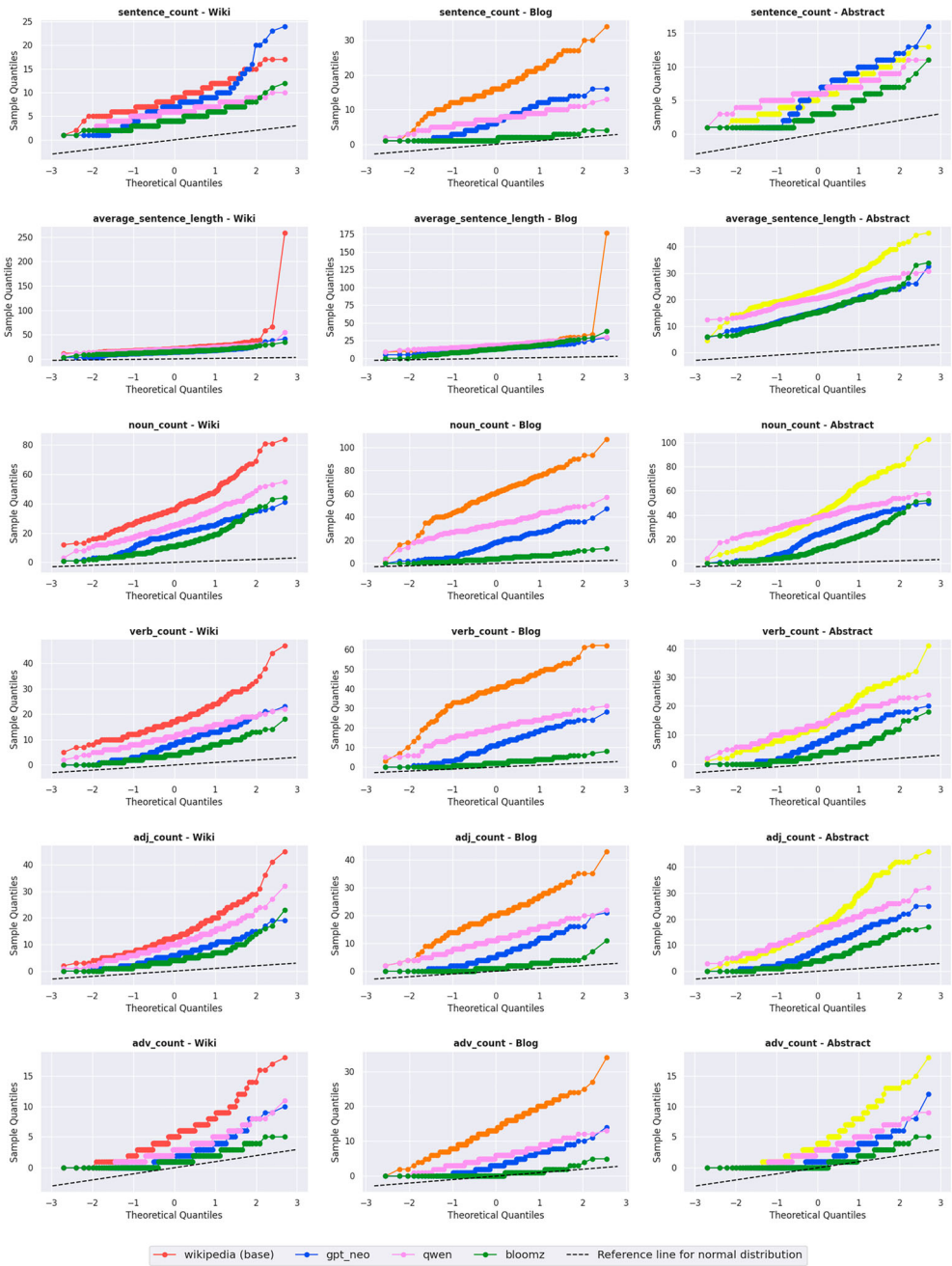


Figure 15. Q-Q plot displaying the distribution of syntactic features across machine-generated and human-authored texts.

normality. In contrast, GPT-neo and Bloomz exhibit significant non-normality across most metrics.

As shown in Figure 19, most features deviate significantly from normality. However, for blog and Qwen intros, features like average word count, reading score or function word percentage show higher p -values, suggesting normality.

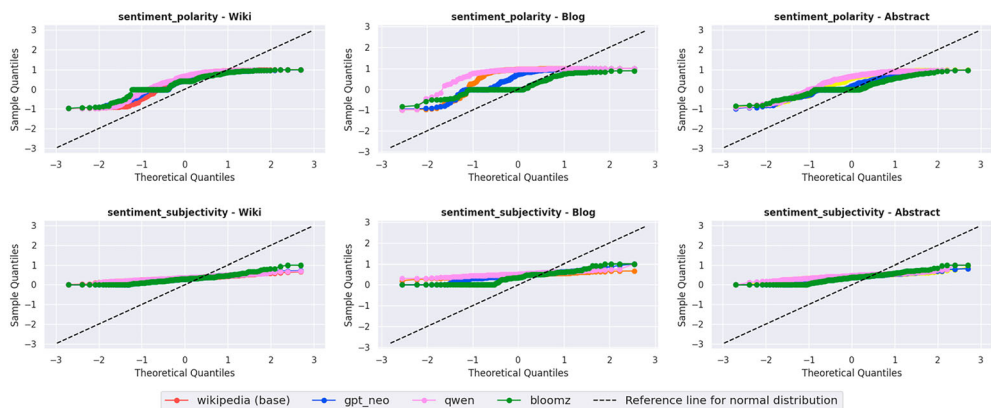


Figure 16. Q-Q plot displaying the sentiment polarity and subjectivity across machine-generated and human-authored texts.

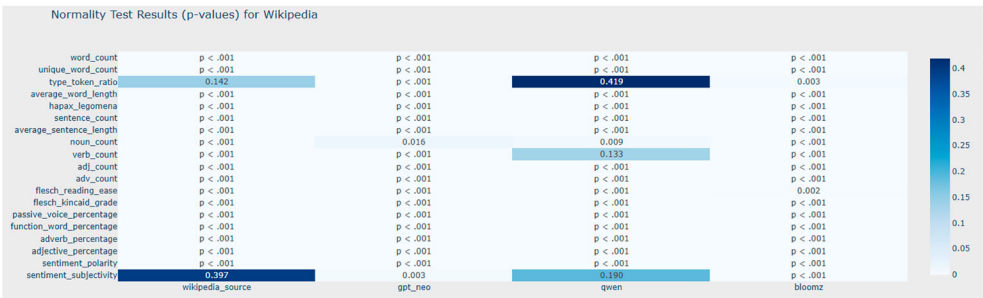


Figure 17. Normality test (p -values) using Shapiro-Wilk Test across machine-generated and Wikipedia texts.

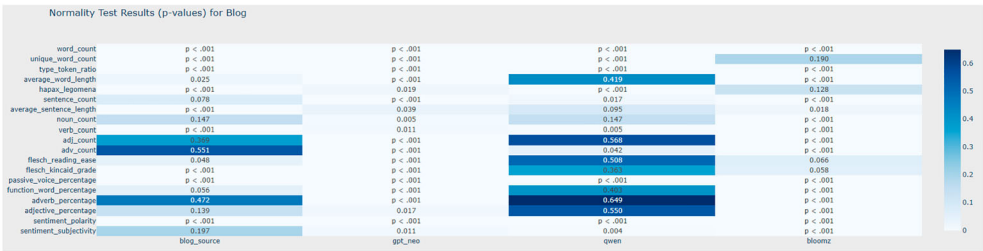


Figure 18. Normality test (p -values) using Shapiro-Wilk Test across machine-generated and blog texts.

While some features in specific datasets meet the criteria for normality, we will use a non-parametric test for further analysis to minimise statistical inaccuracies.

4.5. Group comparison

Because our data is mostly non-normally distributed, we employ the Kruskal – Wallis test to assess whether statistically significant differences exist among the groups. To complement

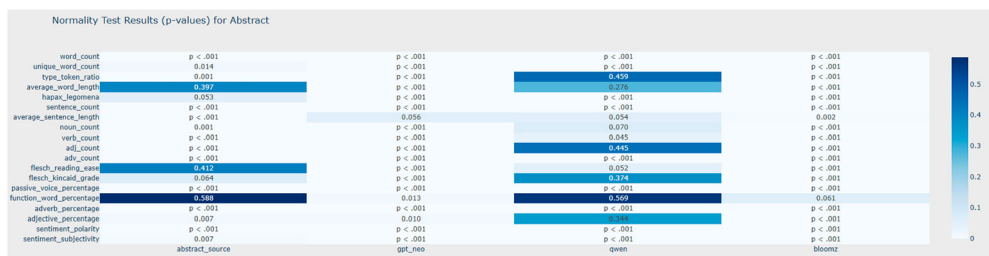


Figure 19. Normality test (p -values) using Shapiro-Wilk Test across machine-generated and abstract texts.

this analysis, we calculate epsilon squared as a measure of effect size, providing insight into the magnitude of these differences. If the Kruskal – Wallis test yields a significant result, we proceed with Dunn’s post-hoc test, applying the Bonferroni correction to account for multiple comparisons and ensure statistical validity.

4.5.1. Group comparison for lexical features

The results in Table 3 reinforce the previous findings, highlighting significant differences between human-generated and machine-generated texts. Dunn’s post-hoc comparisons revealed that nearly every pair of text types showed significant differences across all assessed lexical features. These results emphasise the substantial and systematic variations in the lexical profiles of texts written by humans compared to each language model, as well as differences among the models themselves.

4.5.2. Group comparison for readability metrics

The results shown in Table 4 indicate highly significant differences ($p < 0.001$) in readability metrics across all three groups and between the texts generated by the models. The Kruskal–Wallis values are significant, and the associated effect sizes range from weak to moderate. Dunn’s post hoc tests revealed that nearly every pair of texts differ statistically. However, the results of the abstract source do not show statistical significance for the Qwen model. Additionally, the texts generated by GPT Neo and Bloomz are not statistically different concerning the Wikipedia and blog sources, indicating that their generated texts are highly similar, which aligns with the previous data from the visualisations.

4.5.3. Group comparison for syntactic features

The results presented in Table 5 indicate highly significant differences between the groups, as shown by the p -value < 0.001 from Dunn’s post-hoc test. However, there are instances where the results are not statistically significant. This suggests that for certain features, the generated text does not differ from the source text, or that the models produce similar outputs for the analyzed attributes. The model Qwen shows no statistical differences compared to the Wikipedia source concerning average sentence length, nor does it differ from the abstract source in terms of noun, verb, and adjective counts. Additionally, there is no statistical difference between the model Qwen and GPT regarding sentence count for both Wikipedia and abstract sources.

Table 3. Kruskal – Wallis test, epsilon squared and Dunn’s post-hoc test for group comparison between lexical features of human and machine generated texts.

S	Feature	H	p	ϵ^2	Dunn’s post-hoc test					
					S vs GPT	S vs QWEN	S vs Bloomz	GPT vs QWEN	GPT vs Bloomz	QWEN vs Bloomz
W	word_count	538,65	0,00	0,67	***	***	***	**	***	***
W	unique_word_count	586,64	0,00	0,73	***	***	***	***	***	***
W	type_token_ratio	349,31	0,00	0,44	***	***	***	***	ns	***
W	average_word_length	278,09	0,00	0,35	***	***	***	***	ns	***
W	hapax_legomena	573,27	0,00	0,72	***	ns	***	***	***	***
B	word_count	413,15	0,00	0,81	***	***	***	***	***	***
B	unique_word_count	438,49	0,00	0,86	***	***	***	***	***	***
B	type_token_ratio	292,79	0,00	0,57	**	***	***	***	***	**
B	average_word_length	183,79	0,00	0,36	***	***	***	***	ns	***
B	hapax_legomena	429,97	0,00	0,84	***	***	***	***	***	***
A	word_count	339,64	0,00	0,42	***	ns	***	***	***	***
A	unique_word_count	472,76	0,00	0,59	***	***	***	***	***	***
A	type_token_ratio	143,46	0,00	0,18	ns	***	***	***	***	*
A	average_word_length	90,77	0,00	0,11	***	ns	ns	***	***	***
A	hapax_legomena	524,28	0,00	0,65	***	***	***	***	***	***

S = source, W = Wikipedia, B = blog, A = abstract, ns = non-significant, * = $p < 0,05$, ** = $p < 0,01$, *** = $p < 0,001$

Table 4. Kruskal – Wallis test, epsilon squared and Dunn’s post-hoc test for group comparison between readability metrics of human and machine generated texts.

S	Feature	H	p	ϵ^2	Dunn’s post-hoc test					
					S vs GPT	S vs QWEN	S vs Bloomz	GPT vs QWEN	GPT vs Bloomz	QWEN vs Bloomz
W	flesch_reading_ease	241,76	0,00	0,30	***	*	***	***	ns	***
W	flesch_kincaid_grade	304,99	0,00	0,38	***	ns	***	***	ns	***
B	flesch_reading_ease	153,91	0,00	0,30	***	***	***	***	ns	***
B	flesch_kincaid_grade	166,13	0,00	0,32	***	***	***	***	ns	***
A	flesch_reading_ease	92,47	0,00	0,11	***	ns	***	***	**	***
A	flesch_kincaid_grade	156,95	0,00	0,19	***	ns	***	***	*	***

S = source, W = Wikipedia, B = blog, A = abstract, ns = non-significant, * = $p < 0,05$, ** = $p < 0,01$, *** = $p < 0,001$

Table 5. Kruskal – Wallis test, epsilon squared and Dunn’s post-hoc test for group comparison between syntactic features of human and machine generated texts.

S	Feature	H	p	ϵ^2	Dunn’s post-hoc test					
					S vs GPT	S vs QWEN	S vs Bloomz	GPT vs QWEN	GPT vs Bloomz	QWEN vs Bloomz
W	sentence_count	309,54	0,00	0,39	***	***	***	***	***	***
W	average_sentence_length	294,23	0,00	0,37	***	ns	***	***	ns	***
W	noun_count	411,31	0,00	0,51	***	***	***	***	***	***
W	verb_count	435,76	0,00	0,54	***	***	***	***	***	***
W	adj_count	335,60	0,00	0,42	***	**	***	***	***	***
W	adv_count	269,87	0,00	0,34	***	***	***	***	**	***
B	sentence_count	367,39	0,00	0,72	***	***	***	ns	***	***
B	average_sentence_length	107,45	0,00	0,21	***	ns	***	***	ns	***
B	noun_count	411,55	0,00	0,80	***	***	***	***	***	***
B	verb_count	410,06	0,00	0,80	***	***	***	***	***	***
B	adj_count	368,70	0,00	0,72	***	***	***	***	***	***
B	adv_count	349,34	0,00	0,68	***	***	***	***	***	***
A	sentence_count	190,37	0,00	0,24	ns	*	***	ns	***	***
A	average_sentence_length	320,28	0,00	0,40	***	***	***	***	ns	***
A	noun_count	362,40	0,00	0,45	***	ns	***	***	***	***
A	verb_count	382,82	0,00	0,48	***	ns	***	***	***	***
A	adj_count	360,93	0,00	0,45	***	ns	***	***	***	***
A	adv_count	253,91	0,00	0,32	***	*	***	***	***	***

S = source, W = Wikipedia, B = blog, A = abstract, ns = non-significant, * = $p < 0,05$, ** = $p < 0,01$ *** = $p < 0,001$

4.5.4. Group comparison for stylistic features

The results presented in Table 6 indicate differences among the groups. Dunn's post hoc test reveals significant differences, although some groups show non-significant differences in certain stylistic features. Additionally, the epsilon squared values suggest a weak to moderate effect size. Notably, the Qwen model generates outputs similar to human-written content, especially in the wiki and abstract introductions. Furthermore, the GPT Neo and Bloomz models produce similar text across nearly all features and sources.

4.5.5. Group comparison for sentiment analysis

The results in Table 7 suggest fewer statistically significant differences between machine-generated and human-written text across each group, thereby supporting the previous findings. When comparing the text produced by different models, only GPT-Neo and BLOOMZ showed no statistical differences in the contexts of Wikipedia and abstract sources, suggesting that these models create text that is more similar to one another.

5. Error analysis

In this section, we identify common discrepancies in machine-generated text, analyze their causes and suggest targeted improvements.

Our data shows that Wikipedia introductions typically contain more than 120 words and 70–80 unique words. In contrast, many AI-generated examples remain under 80 words and contain fewer unique terms. Stylistically, they often use too much passive voice, deviating from Wikipedia's more balanced approach. They also differ in tone: while Wikipedia maintains a slightly positive but largely factual style, some models remain overly neutral or lean towards a positive tone, losing the objectivity found in encyclopedia entries.

Comparisons with blog introductions reveal patterns such as an inconsistent tone, inconsistent descriptions and limited structural variety. Blogs tend to exhibit moderate positivity (subjectivity of 0.4–0.6) and a wider range of sentence lengths, but AI outputs are often in a narrower distribution. They also fluctuate between an overly use of verbs and an overly sparse use of adjectives and adverbs, resulting in passages that are either bloated or bland.

Academic abstracts are the most formal and complex among the three text groups, characterised by dense technical vocabulary, intricate sentence structures and a reliance on passive voice and function words. These features contribute to higher lexical diversity and specialised terminology, reflecting the precision required in academic communication. In contrast, machine-generated abstracts exhibit significant simplification. The machine-produced texts tend to have shorter sentences, lower lexical diversity and fewer technical terms, resulting in higher readability and lower grade-level scores.

These discrepancies could be due to the limitation of the training data, insufficient detail in the prompt, and simplification that reduces textual depth and descriptive richness. These results reflect the model's pre-trained nature. While Qwen is an instruction-tuned variant designed to generate coherent and contextually appropriate text (Yang et al., 2024), BloomZ-560M model has versatility in handling diverse linguistic inputs makes it accessible to a broader global audience, removing language barriers that might exist with other models.

Table 6. Kruskal – Wallis test, epsilon squared and Dunn's post-hoc test for group comparison between stylistic features of human and machine generated texts.

S	Feature	H	<i>p</i>	ϵ^2	Dunn's post-hoc test					
					S vs GPT	S vs QWEN	S vs Bloomz	GPT vs QWEN	GPT vs Bloomz	QWEN vs Bloomz
W	passive_voice_percentage	16,42	0,00	0,02	***	ns	ns	*	ns	ns
W	function_word_percentage	138,85	0,00	0,17	***	ns	***	***	ns	***
W	adverb_percentage	69,14	0,00	0,08	***	ns	***	***	ns	***
W	adjective_percentage	74,28	0,00	0,09	ns	***	ns	***	ns	***
B	passive_voice_percentage	29,43	0,00	0,05	*	ns	***	ns	ns	**
B	function_word_percentage	88,48	0,00	0,17	***	***	ns	***	**	***
B	adverb_percentage	63,31	0,00	0,12	***	*	***	ns	**	***
B	adjective_percentage	35,17	0,00	0,06	ns	**	ns	***	ns	***
A	passive_voice_percentage	28,21	0,00	0,03	*	ns	***	*	ns	***
A	function_word_percentage	59,00	0,00	0,07	***	ns	ns	***	***	ns
A	adverb_percentage	127,24	0,00	0,16	***	**	***	***	ns	***
A	adjective_percentage	46,75	0,00	0,05	***	ns	***	***	ns	*

S = source, W = Wikipedia, B = blog, A = abstract, ns = non-significant, * = $p < 0,05$, ** = $p < 0,01$ *** = $p < 0,001$

Table 7. Kruskal – Wallis test, epsilon squared and Dunn's post-hoc test for group comparison between sentiment neutrality and subjectivity of human and machine generated texts.

S	Feature	H	<i>p</i>	ϵ^2	Dunn's post-hoc test					
					S vs GPT	S vs QWEN	S vs Bloomz	GPT vs QWEN	GPT vs Bloomz	QWEN vs Bloomz
W	sentiment_polarity	12,42	0,01	0,01	ns	ns	ns	*	ns	**
W	sentiment_subjectivity	20,85	0,00	0,02	ns	ns	*	**	ns	***
B	sentiment_polarity	162,43	0,00	0,31	***	ns	***	***	***	***
B	sentiment_subjectivity	39,58	0,00	0,07	ns	*	**	*	*	***
A	sentiment_polarity	93,58	0,00	0,11	***	ns	***	***	ns	***
A	sentiment_subjectivity	36,11	0,00	0,04	ns	ns	**	***	ns	***

S = source, W = Wikipedia, B = blog, A = abstract, ns = non-significant, * = $p < 0,05$, ** = $p < 0,01$ *** = $p < 0,001$

In addition, the generated text often relies on repetitive patterns and clichés, which reduces originality and engagement. Studies on the creative capabilities of LLMs show that open-source models are particularly prone to the reuse of common phrases, which limits their attractiveness in creative domains (Hayawi et al., 2023).

To address these shortcomings, domain-specific fine-tuning can help models learn the structures and styles needed for diverse writing contexts. Iterative feedback – combining linguistic metrics with editorial reviews – can further refine text features accuracy, while personalised prompts can enhance overall performance.

6. Discussion

This study examines the distinct characteristics of human-authored and machine-generated introductions, focusing on three text categories: Wikipedia introductions, Medium blog article introductions, and ArXiv scientific abstracts. Each category is anchored to a human-authored reference, with corresponding machine-generated texts produced using three selected generative models. The analysis is structured around five key linguistic dimensions: lexical features, syntactic structures, readability metrics, stylistic elements, and sentiment analysis. Despite the topics of each text source (*Appendix 1*), the differences generally align with existing literature and highlight both the capabilities and limitations of current open-source AI models.

Our results consistently demonstrated that AI-generated text exhibits lower lexical diversity, confirming previous findings (Ma et al., 2023; Reviriego et al., 2023). For instance, GPT-Neo and BloomZ produced fewer unique words and had a lower TTR compared to human-authored texts. In contrast, Wikipedia, blog and scientific content exhibited a larger word count and more diverse syntactic patterns, reflecting the authors' ability to incorporate personal experiences, emotional undertones, and stylistic subtleties (Babiker, 2023; Jones, 2022; Kojima, 2024; Maamujav et al., 2021; Verma & Srinivasan, 2019). This creativity and diversity of expression emphasise the value of human authorship (Grimaldi & Ehrler, 2023; Ma et al., 2023).

Similar to prior work (Imperial & Madabushi, 2023), our analysis revealed that AI-generated texts often feature simpler sentence structures and higher readability, as reflected in their higher Flesch Reading Ease scores. Models like GPT-Neo and BloomZ consistently demonstrate more accessible language, relying on shorter sentences and simple clauses aligning with studies highlighting generative models' tendencies to produce text with reduced syntactic depth (Leong, 2023; Liu et al., 2023).

While following the given prompt instructions, the models demonstrated varying levels of effectiveness in replicating the target writing style. BloomZ produced text characterised by conciseness and repetitiveness, whereas GPT-Neo exhibited intermediate stylistic features. Among the models, Qwen demonstrated the closest approximation to the intended writing style (Jones, 2022; Kojima, 2024; Muñoz-Ortiz et al., 2023).

Recent progress in generative AI has opened up new possibilities in many areas, including education, gaming, advertising, painting, and coding (Zhang et al., 2023). Our observation of high clarity and consistency in machine-generated introductions aligns with this progress, suggesting that AI can be particularly useful for certain tasks, such as generating summaries, or short headlines or in chatbots where straightforward language is needed. However, there are still existing concerns. AI struggles with providing reliable facts and

expressing ideas with depth. Various studies (Grimaldi & Ehrler, 2023; Herbold et al., 2023; Ma et al., 2023) note that AI outputs can include inaccuracies, repetition, or contradictory statements – an observation reinforced by our finding that machine-generated texts have narrower distributions and simpler linguistic variety, implying reliance on statistical patterns rather than deep subject-matter understanding. This could become a problem in medical, legal, and educational domains, for example, where mistakes or oversimplifications of text could be misleading (Cheng et al., 2023; Herbold et al., 2023; Ma et al., 2023).

Although, the performance of our selection in AI models may differ significantly from more advanced or fine-tuned models, especially in generating domain-specific content. AI models explicitly trained on specialised datasets or fine-tuned for particular tasks better address the complexities and specificities of their respective domains. These specialised models can achieve higher accuracy and relevance by leveraging their domain-focused training. However, human fact-checking and adding appropriate disclaimers are necessary, especially in situations where precision and accuracy are very important (Ma et al., 2023).

In our study, we faced some limitations. First, our focus on open-source models, chosen for their accessibility, transparency, and reproducibility, does not capture the entire landscape of AI technologies. The differences found in the literature suggest that some open-source models with appropriate fine-tuning can compete with or even outperform proprietary systems (e.g. GPT-3.5, GPT-4, Claude) on certain tasks – WizardMath, for example, outperformed GPT-3.5 and Claude on mathematical reasoning (Luo et al., 2025). GPT-Neo and BloomZ produced nearly identical responses to repeated prompts (*Appendix 2*), while Qwen offered slight variations, suggesting a more limited generative capacity compared to larger proprietary models (An et al., 2023; Wu et al., 2023). Also, for our analysis, we utilised a reference dataset comprising 200 Wikipedia articles to facilitate comparisons with an encyclopedic and formal writing style, 128 Medium blog articles to assess alignment with a more natural and descriptive style, and 200 ArXiv scientific abstracts to examine the capacity of models to generate technical and specialised texts. In total, the study analyzed 2112 texts, 528 being the human-authored texts. While that was enough to start making comparisons, it might not fully show all the different topics and writing styles. Additionally, we only focused on the introductions of articles, which can follow a certain structure and style. Finally, we didn't fine-tune the AI models for specific tasks, so their performance might change if they were customised for particular purposes.

Future research should explore a broader range of AI models and employ domain-specific fine-tuning to further close the gap between machine and human-generated content and account for stylistic errors.

Overall, this study highlights both the advancements and the ongoing challenges of machine-generated writing. These insights can guide improvements in AI models, enabling them to better replicate the depth and richness of human expression. Additionally, understanding these differences can support the effective integration of AI tools across various domains, ensuring that the generated content meets the desired standards of quality and depth.

7. Conclusion

This research presents a detailed framework for comparing text properties of machine-generated and human-authored texts. Using three openly available generative models,

GPT-Neo 1.3B, Qwen2.5-1.5B-Instruct, and BloomZ-560M from Hugging Face, we systematically analyzed their outputs alongside human-written Wikipedia and Medium blog introductions and arXiv scientific abstracts. Descriptive statistics and non-parametric tests revealed notable distinctions in text features between machine-generated and human-authored samples. Although the models demonstrated significant progress in generating coherent and readable content, they still lacked the depth, nuance and stylistic richness that characterise human texts.

The observed differences in terms of lexical richness, syntactic diversity and sentiment polarity are likely due to the unique training procedures and underlying architectures of each model. GPT-Neo, Qwen and BloomZ rely on different corpora, parameter sizes and fine-tuning techniques that influence the language patterns they capture.

For example, GPT-Neo often produced concise, easy-to-read text with moderately neutral to positive sentiment, while BloomZ often produced simpler, shorter texts. Qwen showed some overlap with human texts on several metrics, but did not achieve the full range of stylistic and lexical nuances found in human-authored texts.

Although our results highlight the potential of AI-generated content in improving accessibility and efficiency, e.g. how these selected models can simplify complex concepts and make them accessible to a wider audience, they also highlight the need for further progress to address gaps in lexical richness, syntactic complexity and stylistic nuance.

CRediT authorship contribution statement

Lia Cornelia Culda: conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing. **Raluca Andreea Nerişanu:** conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing. **Marian Pompiliu Cristescu:** conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing, supervision. **Dumitru Alexandru Mara:** conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing. **Adela Bâra:** conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing, supervision. **Simona Oprea:** conceptualisation, methodology, software, formal analysis, writing – original draft, writing – reviewing and editing, supervision.

Data availability

We used publicly available data. Original data set: <https://doi.org/10.57967/hf/0326>, <https://huggingface.co/datasets/fabiochui/medium-articles> and <https://huggingface.co/datasets/gfissore/arxiv-abstracts-2021>.

Generativ models Transformers:

- <https://huggingface.co/EleutherAI/gpt-neo-1.3B?library=transformers>
- <https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct?library=transformers>
- <https://huggingface.co/bigscience/bloomz-560m?library=transformers>

Author contributions

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analysis, Methodology, Software, Writing – original draft, Writing – review & editing; **Marian Pompiliu Cristescu**: Conceptualization, Formal analysis, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing; **Dumitru Alexandru Mara**: Conceptualization, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing; **Adela Bâra**: Conceptualization, Formal analysis, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing; **Simona-Vasilica Oprea**: Conceptualization, Formal analysis, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing

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Appendices

Appendix 1. Summarisation of covered topics.

Source	Total number of articles	Categories	Number of articles per category
Wikipedia	200 articles	Historical places / Cities / Towns / Vehicles / Streets	27
		Technology / Engineering / Business	14
		Entertainment (tv, movies, series, awards, radio, games, news)	25
		Literature	8
		Biographies	81
		Styles / Cultures	13
		Animals / Plants / Foods	10
		Language / Law	4
		Universe / Nature	10
		Medicine / Science	8
		Technology & Data (Tags: Science, Neuroscience, Artificial Intelligence, Technology, Analytics, Data Science, Exploratory Data Analysis, Data Engineering, Business Intelligence, Programming, UX, Software Engineering, React, Machine Learning, Python, Engineering)	37
		Lifestyle & Wellness (Tags: Lifestyle, Motivation, Life, Storytelling, Work, Life Lessons, Wellness, Weight Loss, Fitness Tips)	20
Blog	128 articles	Climate & Sustainability (Tags: Climate Change, Sustainability, Environment)	7
		Personal Growth & Arts (Tags: Poetry, Music, Books, Mental Health, Self Improvement)	16
		Writing & Media (Tags: News, Journalism, Writing, Authenticity)	19
		Business & Innovation (Tags: Marketing, Creativity, Entrepreneurship, Start-up, Freelancing, Startup, Blockchain)	29
Scientific abstracts	200 articles	Astrophysics / Cosmology	55
		Mathematics and Statistics	66
		Computer science	8
		Physics/ Nuclear theory/ Quantum energy	71

Appendix 2. Generated outputs before cleaning procedures.

Wikipedia introduction	
KGHP (89.9 FM) is a student-run high school radio station operating on a non-commercial license in Gig Harbour, Washington. Owned by the Peninsula School District #401, the station's studio is located on the campus of Peninsula High School. With its two translators, K207AZ 89.3 and K289BZ 105.7, the station's signal covers most of the Gig Harbour Peninsula, Key Peninsula and portions of Tacoma, Fox Island, Washington and Olympia, Washington. The station came on the air in 1988 and was one of three high school radio station in the state of Washington. The first manager and teacher was Don Hofmann, a former KNBQ-FM general manager. KSTW-TV technician Max Bice was the engineer. The current station manager is Spencer "Walrus" Abersolds. The multi-format station is run by students at Peninsula High School and Gig Harbour High School during the day, and in the evening to night hours it is run by community volunteers. The station gives a variety of shows from the students which range from sports broadcasts to classic rock.	
GPT_neo_intro	
First run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high school radio station in the San Francisco Bay Area. It is the only high-school radio station in the Bay area that is not affiliated with a college or university. KGGL (88.7 FM), a non-commercial student radio and television station in the San Jose, California area, is owned by the University of California. KGGL is an affiliate of the UC radio network. The station's programming is broadcast from the campus of San José State University. The following is from a letter from K.G.H.P. to the editor of The San Francisco Chronicle. It was written by a member of KGHP's student staff. I have not been able to find any other information about the student staff. If anyone has any information,
Second run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high school radio station in the San Francisco Bay Area. It is the only high-school radio station in the Bay area that is not affiliated with a college or university. KGGL (88.7 FM), a non-commercial student radio and television station in the San Jose, California area, is owned by the University of California. KGGL is an affiliate of the UC radio network. The station's programming is broadcast from the campus of San José State University. The following is from a letter from K.G.H.P. to the editor of The San Francisco Chronicle. It was written by a member of KGHP's student staff. I have not been able to find any other information about the student staff. If anyone has any information,
Qwen_intro	
First run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high-power radio station in New Haven, Connecticut, United States. It broadcasts from the University of Connecticut campus at 1350 West Eagleville Road. The station's format is rock and pop music with some news programming. In its first year as an operating station, KGHPR received several awards including "Best Student Radio Station" by the National Association of Broadcasters and "Outstanding Program for High School Students" from U.S. News & World Report. In 27 years of operation, it has won more than 60 awards, most notably winning three consecutive "Student Programming Awards" in 4KU's annual competition. The original name of the station was KJMP, which stood for "Just Jumping On the Media Bandwagon". This acronym was later changed to KGHM, standing for K-Great-H
Second run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high-power radio station located in the University of Pittsburgh's Oakland campus. It broadcasts educational programming, news, and music to the local community. The station has been active since 1965 and has won several awards for its programming. The radio format at KGHS varies throughout the year based on events such as academic conferences, cultural festivals, sports games, etc. Programming includes talk shows, interviews with students, faculty members, guest artists, live performances, classical music, rock & roll, jazz, blues, hip hop, electronic dance music (EDM), and other genres. In addition to its main broadcast, KGPS also offers satellite internet services and provides free Wi-Fi access to students. This service allows them to connect their devices directly to KGSHS' server via a wired connection or wireless signal. The station's success can be attributed to both its
Bloomz_intro	
First run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high school in the city of St. Louis Missouri USA. KGHS is the name of the school. The school is located in St Louis, Missouri. It is part of a larger school system that includes the St Mary's High School and St John's School.
Second run	200 word wikipedia style introduction on 'KGHP' KGHP (89.9 FM) is a student-run high school in the city of St. Louis Missouri USA. KGHS is the name of the school. The school is located in St Louis, Missouri. It is part of a larger school system that includes the St Mary's High School and St John's School.

Glossary

Feature	Metrics	Definition
Lexical (f_l)	word count – total number of words in text. unique word count – the number of distinct words used. type-token ratio (TTR) – the ratio of unique words to total words, indicating lexical diversity. average word length – The mean length of words in the text reflecting text complexity. hapax legomena – the count of words that appear only once, indicating author originality.	Lexical features refer to measurable characteristics of a text. These features help in distinguishing between different writing styles and recognising linguistic patterns.
Syntactic (f_s)	sentence count – the total number of sentences in the text. average sentence length – the mean number of words per sentence. counts of nouns – the total number of nouns, object-oriented. counts of verbs – the total number of verbs, action-oriented. counts of adjectives – the total number of adjectives, detail-oriented. counts of adverbs – the total number of adverbs, writing modification-oriented.	Syntactic features refer to the structural features of a text and focus on sentence structure and grammatical elements. These features help in analyzing the complexity, style and grammatical variety of a text
Readability (f_r)	Flesch Reading Ease – indicates the ease of reading a text. Flesch-Kincaid Grade Level – indicates the grade level to understand a text, addressing text engagement.	Readability metrics assess the accessibility of text in terms of reading ease or difficulty.
Stylistic (f_v)	passive voice percentage – count the passive constructions used in text. function word percentage – use of grammatical structures (e.g. and, the, in, of, etc.). adverb percentage – measures the proportion of adverb usage. adjective percentage – measures the proportion of adjective usage.	Stylistic features address the writing style, formal, informal, narrative, structured, descriptive, factual or richer and expressive texts.
Sentiment analysis (f_a)	sentiment polarity – sentiment tonality (positive, negative, neutral). sentiment subjectivity – measures how opinion-based (subjective) or fact-based (objective) a text is.	Sentiment analysis examines the emotional tone (positive, neutral or negative) and the subjectivism or objectivism.